

CSA 1718 – Artificial Intelligence

Name : P. Yaswanth

Reg No : 192524167

Prolog Program to Implement Monkey Banana Problem

Aim

To write and implement a Prolog program to solve the Monkey Banana problem using state-space representation.

Objective

- To understand state representation in AI.
- To represent actions as transitions between states.
- To solve a simple problem using Prolog rules.

Theory

The Monkey Banana problem is a classical Artificial Intelligence problem. A monkey needs to obtain a banana placed at a certain position. The monkey can move to the box, climb the box, and obtain the banana.

A state can be represented as:

state(MonkeyPosition, MonkeyStatus, BoxPosition, BananaStatus)

Algorithm

1. Represent the initial state.
2. Define the action of moving to the box.
3. Define the action of climbing the box.
4. Define the action of obtaining the banana.
5. Define the goal state.
6. Recursively search for the goal state.

Program

```
move(state(middle, floor, middle, no),  
      state(middle, box, middle, no)).
```

```
move(state(middle, box, middle, no),  
      state(middle, box, middle, yes)).
```

```
solve(state(middle, box, middle, yes)).
```

solve(State) :-

```
    move(State, NewState),  
    solve(NewState).
```

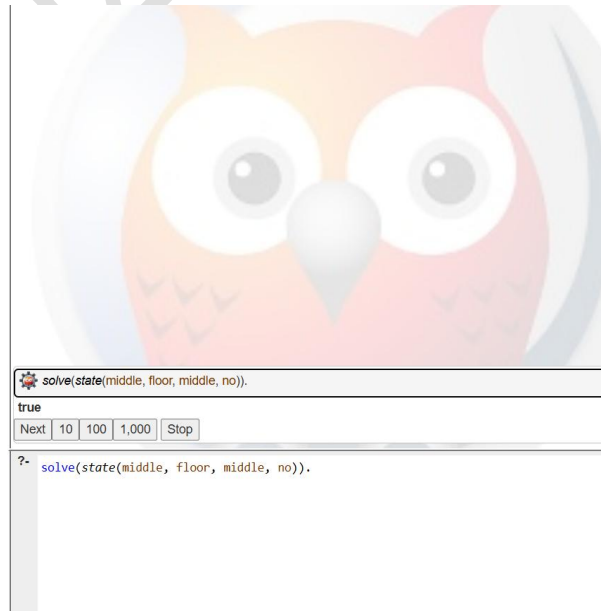
Query

```
solve(state(middle, floor, middle, no)).
```

Output

true.

```
1 move(state(middle, floor, middle, no),  
2   state(middle, box, middle, no)).  
3  
4 move(state(middle, box, middle, no),  
5   state(middle, box, middle, yes)).  
6  
7 solve(State) :-  
8   move(State, NewState),  
9   solve(NewState).  
10  
11 solve(state(middle, box, middle, yes)).
```



Result

Thus, the Monkey Banana problem was successfully represented and solved using Prolog.